

Lab 01

TITLE

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name:

Irza Hameed

Registration number:

(2023-BSCS-44)

Section:

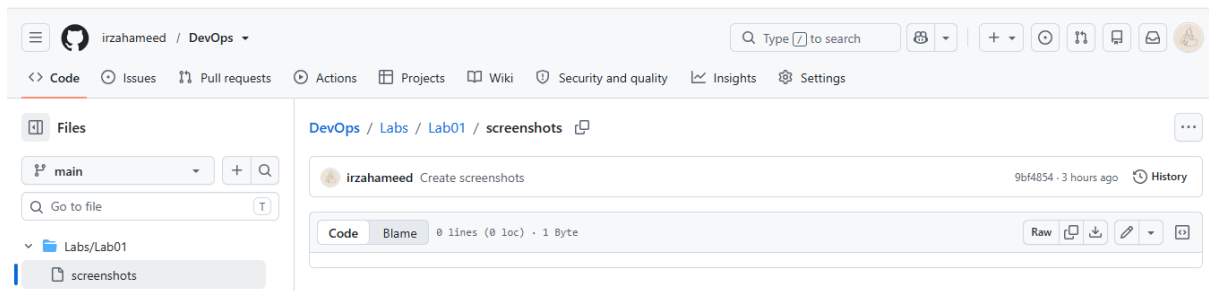
A

GitHub username:

irzahameed

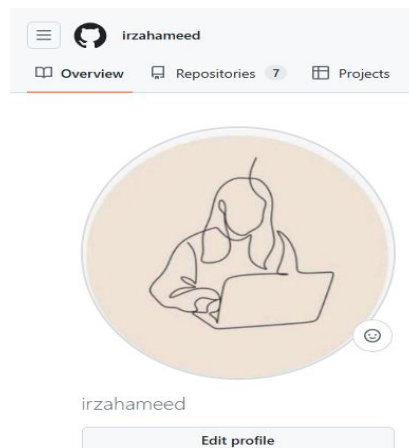
Getting Started

1. Create the following directory inside your DevOps repository:
2. Labs/Lab01/screenshots/
 - o Save a screenshot showing the created directory structure as lab01_folder_structure.png.
3. Read the complete task list before beginning the installation.

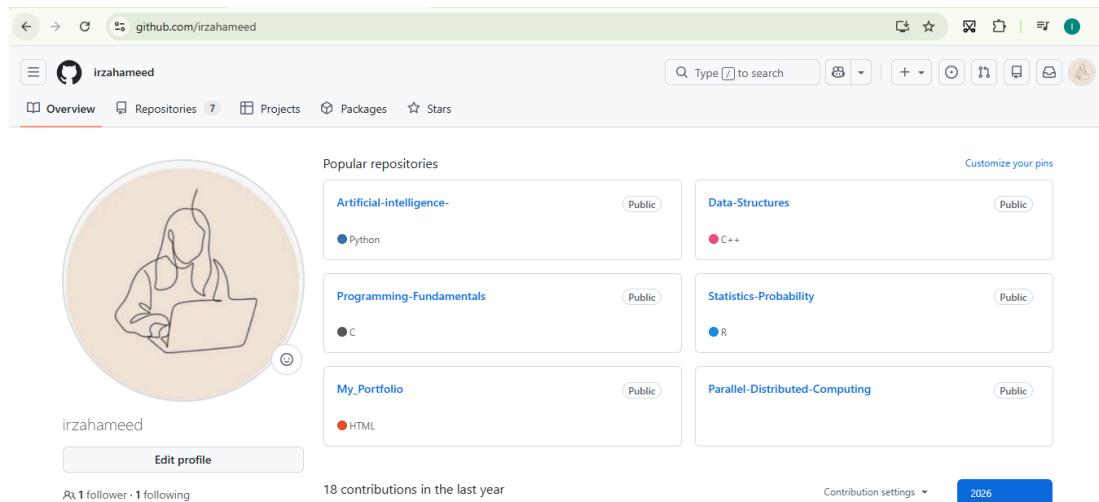


Task 1: Create and Configure Your GitHub Account

1. Open GitHub Sign Up and create your personal GitHub account.
 - Use your own email address and actual identity.
 - Do not create an account using another student's information.
2. Choose a professional GitHub username related to your actual name.
 - Do not use spaces in the GitHub username.
 - Prefer lowercase letters and numbers because the same username will be used in Ubuntu.
 - Save a screenshot clearly showing your profile name and GitHub username as github_profile.png.



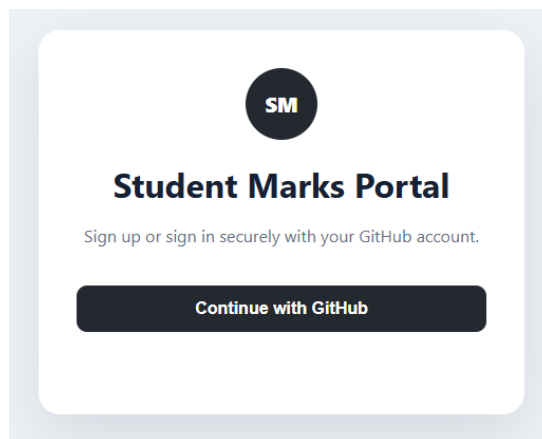
3. Open GitHub > Settings > Public profile.
 - Enter your complete actual name in the Name field.
 - Save a screenshot showing the completed Name field as github_actual_name.png.



- Do not include your password, access token, recovery codes, or other private credentials in any screenshot.

Task 2: Register in the Student Marks Portal

- Open the [Student Marks Portal](#).
- Save a screenshot of the portal login page as portal_login_page.png.



- Select Continue with GitHub and sign in using the GitHub account created in Task
- Enter your actual registration number using the format provided by your instructor, for example:

2024-BSE-00

Select the correct course code.

Select your actual section, either A or B.

Review your information and submit the enrollment for approval.

Save a screenshot showing your GitHub profile information inside the portal as portal_github_profile.png.

Save a screenshot of the submitted or pending enrollment as portal_enrollment_submitted.png.

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2023-BSCS-44

Allowed: 2023-BSCS-01 to 2023-BSCS-99; 2024-BSE-01 to 2024-BSE-99.

Course

CS-649 — Cloud DevOps

Section

Section A

Submit enrollment

Sign out

STUDENT MARKS PORTAL

Sign out

Course enrollments



irzahameed
2023-BSCS-44

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud DevOps (CS-649) Section A
Waiting for administrator approval
Pending approval

STUDENT MARKS PORTAL

Sign out

Academic Record



irzahameed
GitHub authenticated

Registration
2023-BSCS-44

GitHub username
[irzahameed](#)

Email
irzahameed65@gmail.com

Approved courses
1

+ Enroll in another course

APPROVED COURSES

Course Marks

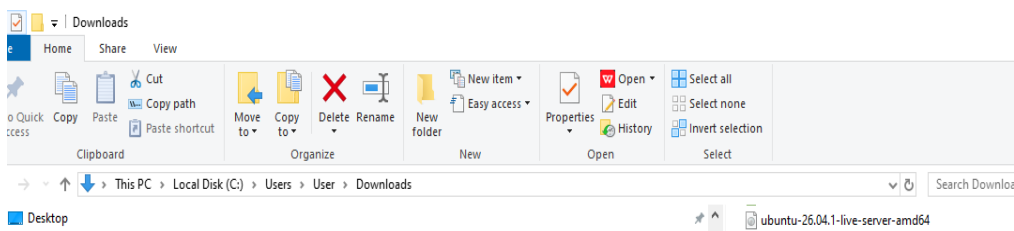
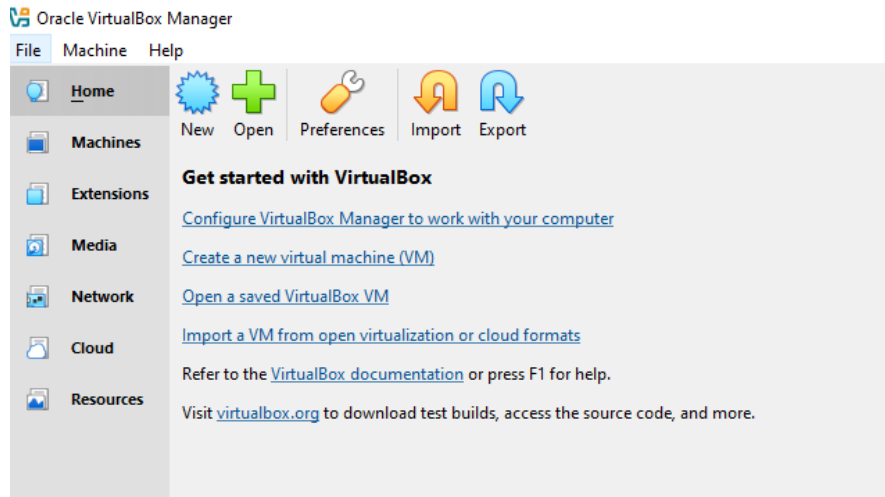
Cloud DevOps (CS-649) Section A

Not published

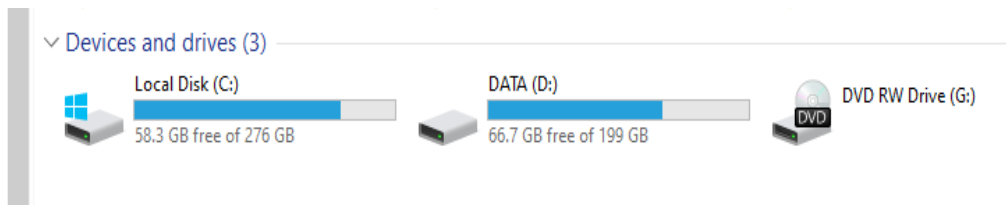
A registration number belongs to only one student. Never enter another student's registration number.

Task 3: Download VMware Workstation Pro and Ubuntu Server

1. Download and install VMware Workstation Pro on your Windows computer.
- If VMware Workstation Pro is already installed, open it and verify that it works.
 - Save a screenshot showing VMware Workstation Pro running as vmware_workstation.png.

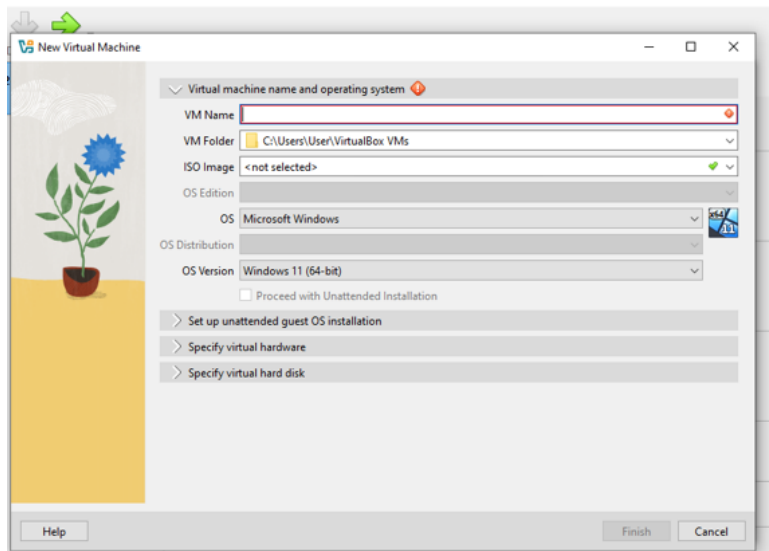


2. Open the Ubuntu Server download page and download the latest Ubuntu Server LTS ISO.
- Save a screenshot showing the downloaded Ubuntu Server ISO filename as ubuntu_server_iso.png.
3. Confirm that at least 20 GB of free storage is available for the Ubuntu Server virtual machine.
- Save a screenshot showing the available storage as available_storage.png.

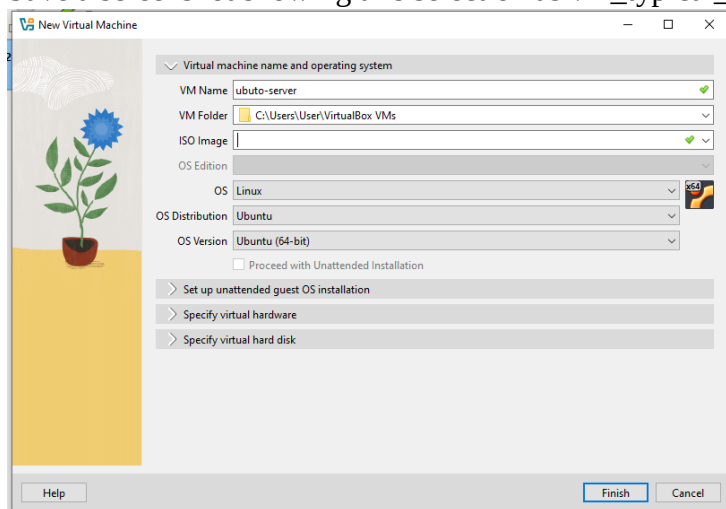


Task 4: Create the Ubuntu Server Virtual Machine

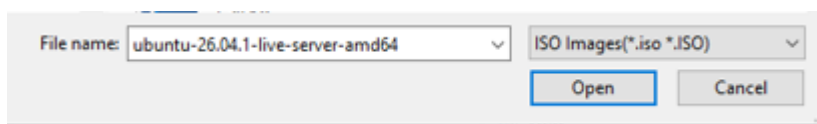
1. Open VMware Workstation Pro and select Create a New Virtual Machine.
- Save a screenshot of the New Virtual Machine Wizard as vm_creation_wizard.png.



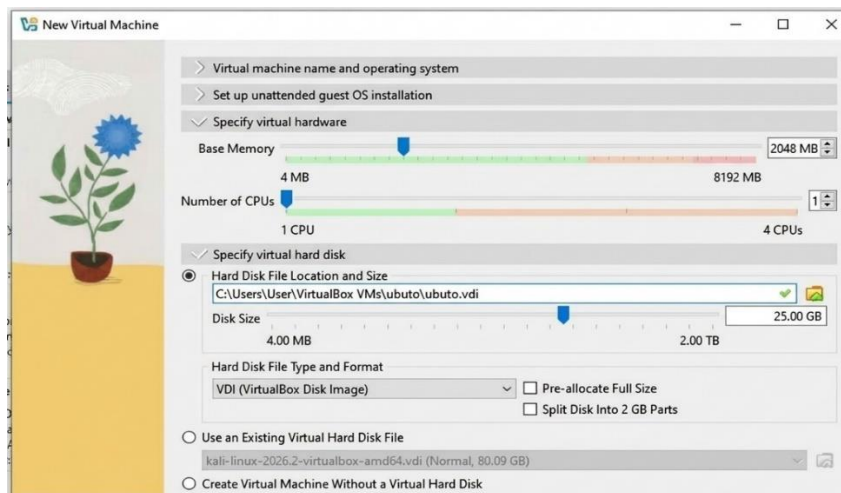
2. Select Typical (recommended) as the virtual machine configuration.
 - Save a screenshot showing this selection as `vm_typical_configuration.png`.



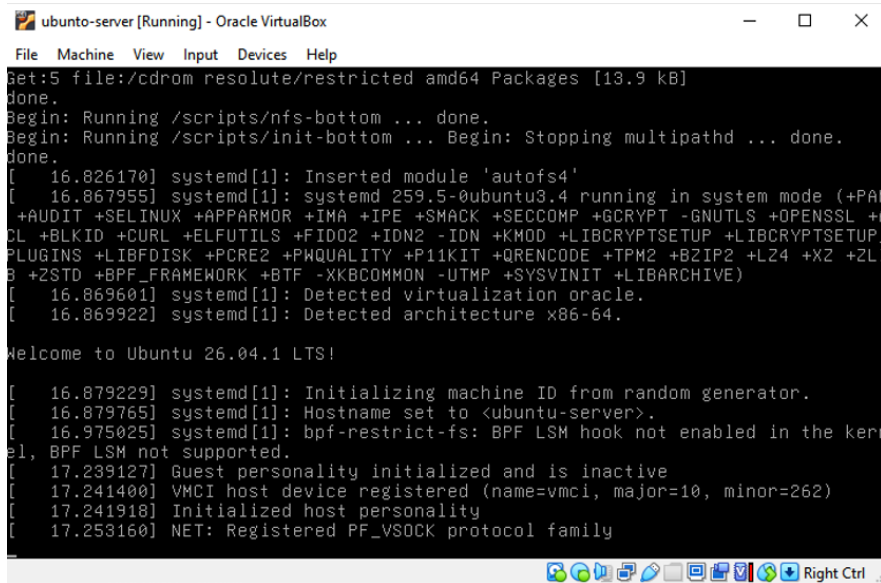
3. Select Installer disc image file (ISO) and browse to the Ubuntu Server ISO downloaded in Task 3.
 - Save a screenshot showing the selected Ubuntu ISO path as `vm_ubuntu_iso_selected.png`.



4. Allocate at least 20 GB of virtual disk space and complete the virtual machine wizard.
 - Save a screenshot showing the final virtual machine configuration as `vm_configuration_summary.png`.



5. Power on the virtual machine and wait for the Ubuntu Server installer to start.
 - Save a screenshot of the Ubuntu Server boot screen as ubuntu_installer_boot.png.



Instructional reference

The following image is provided to help you recognize the Ubuntu Server boot screen:

Task 5: Choose the Installation Language

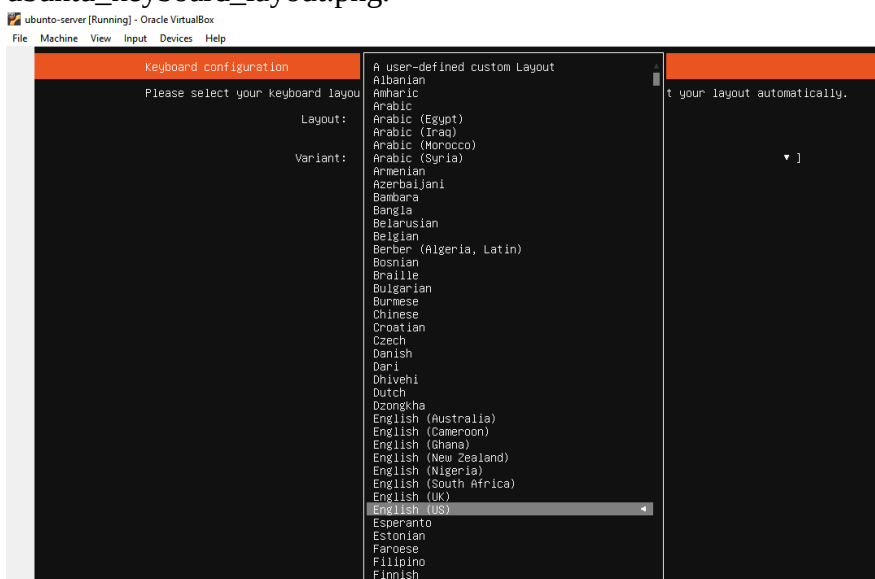
1. Use the arrow keys to select the required installation language.
 - Save a screenshot showing the selected language as ubuntu_language.png.



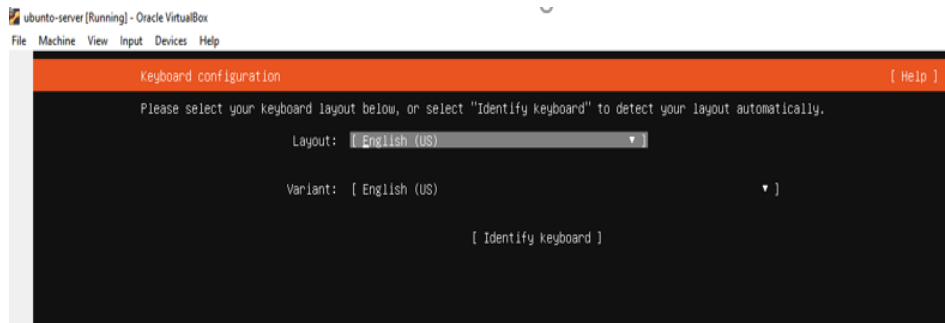
2. Press Enter to continue.

Task 6: Configure the Keyboard Layout

1. Select the keyboard layout that matches your keyboard.
- Save a screenshot showing the selected keyboard layout as `ubuntu_keyboard_layout.png`.



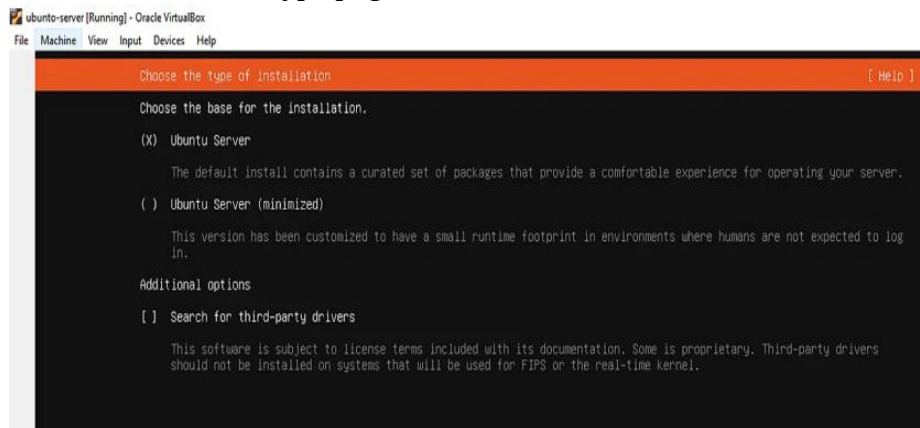
2. Select the appropriate keyboard variant.
- Save a screenshot showing the selected keyboard variant as `ubuntu_keyboard_variant.png`.



3. Select Done to continue.

Task 7: Choose the Ubuntu Server Installation Type

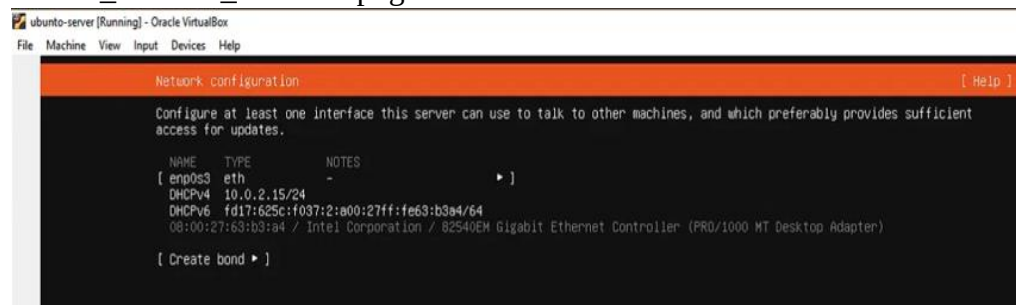
1. Select the normal Ubuntu Server installation option.
 - Do not select a MAAS installation unless your instructor specifically asks for it.
 - Save a screenshot showing the selected installation type as `ubuntu_installation_type.png`.



2. Continue to the next installer screen.

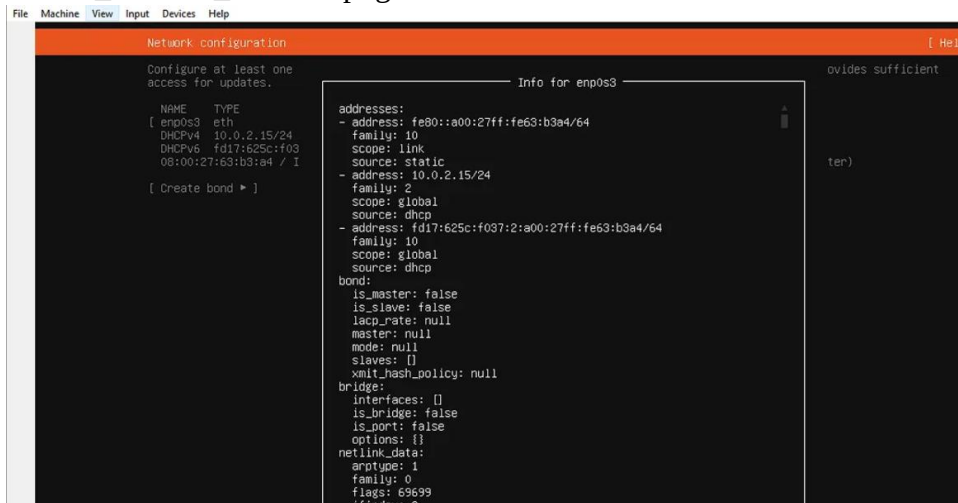
Task 8: Configure the Network

1. Review the network interface detected by the Ubuntu Server installer.
 - Save a screenshot showing the detected network interface as `ubuntu_network_interface.png`.



2. Confirm that the interface has received an IP address automatically through DHCP.

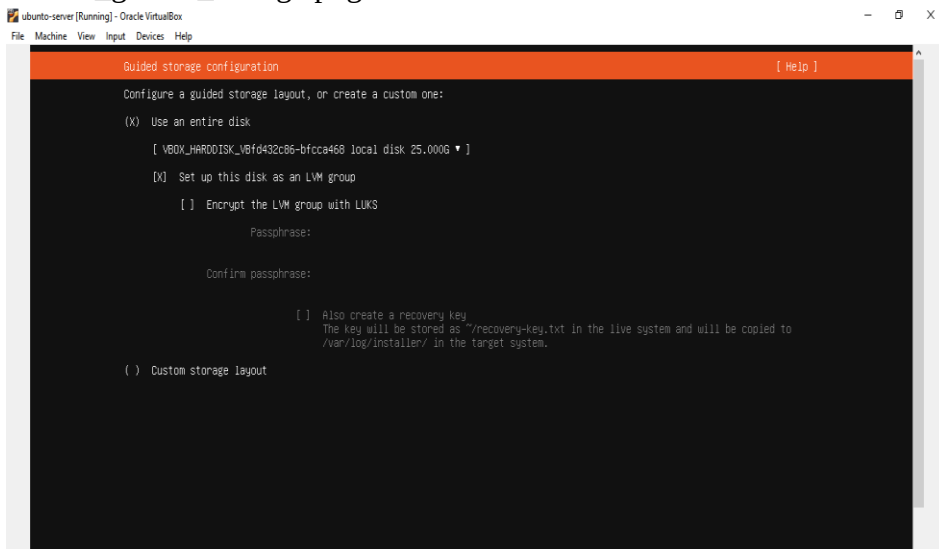
- Save a screenshot showing the assigned network address as ubuntu_installer_network.png.



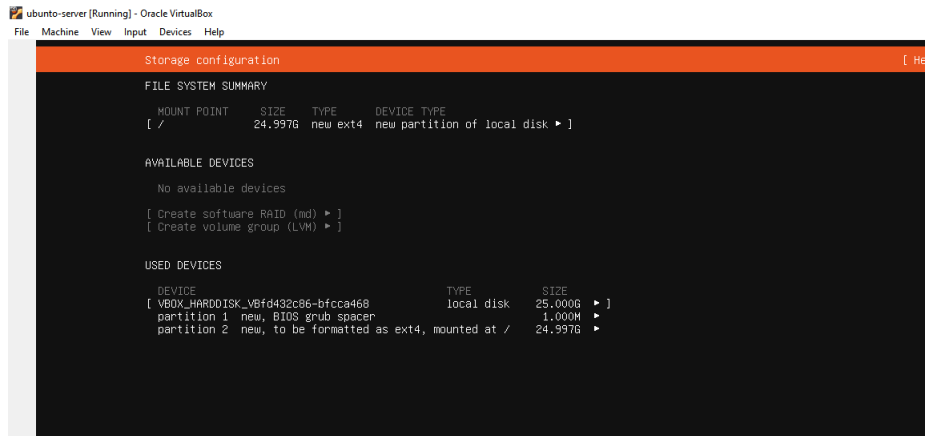
3. Continue when the network configuration is correct.

Task 9: Configure Storage

1. Select the guided storage option to use the VMware virtual disk.
- Save a screenshot showing the selected guided storage option as ubuntu_guided_storage.png.



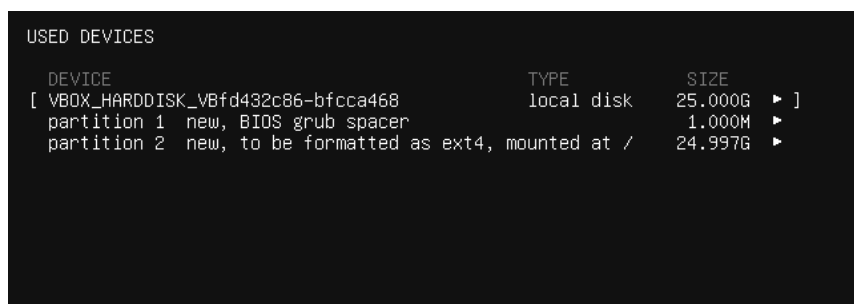
2. Review the storage configuration before continuing.
- Save a screenshot of the storage configuration as ubuntu_storage_configuration.png.



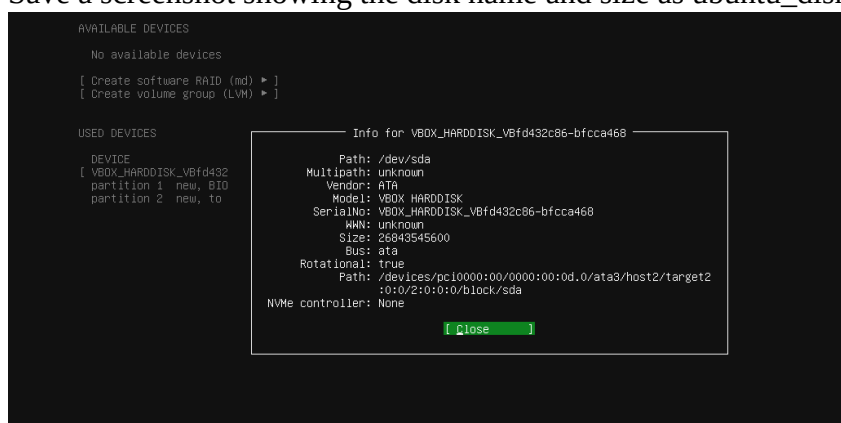
Use only the VMware virtual disk created for this lab. Selecting or formatting the wrong physical disk can destroy existing data.

Task 10: Select the Installation Disk

1. Select the virtual disk created for the Ubuntu Server virtual machine.
 - Save a screenshot showing the selected disk as ubuntu_selected_disk.png.



2. Check the displayed disk size and make sure it is the correct VMware virtual disk.
 - Save a screenshot showing the disk name and size as ubuntu_disk_details.png.



Task 11: Review the Partitions

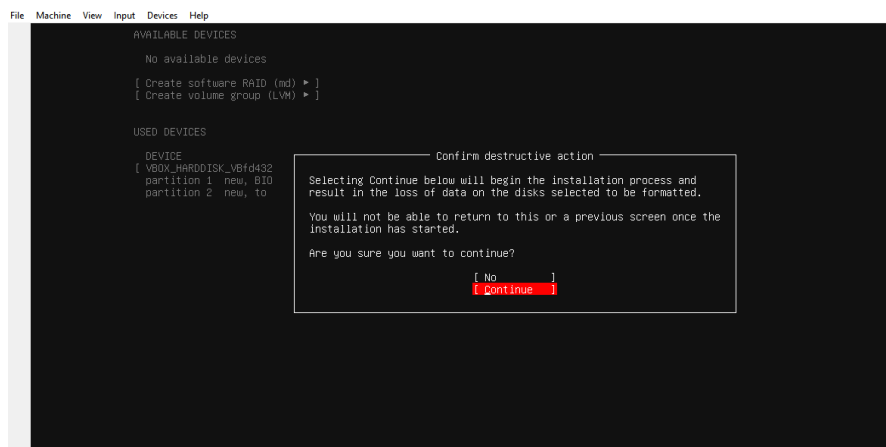
1. Review the partition layout proposed by the Ubuntu Server installer.
 - Save a screenshot showing the proposed partitions as ubuntu_partition_layout.png.

```
[ VBOX_HARDDISK_VBf0432c00-b7cc4988      local disk      25.000G ▶ ]
partition 1  new, BIOS grub spacer          1.000M ▶
partition 2  new, to be formatted as ext4, mounted at /    24.997G ▶
```

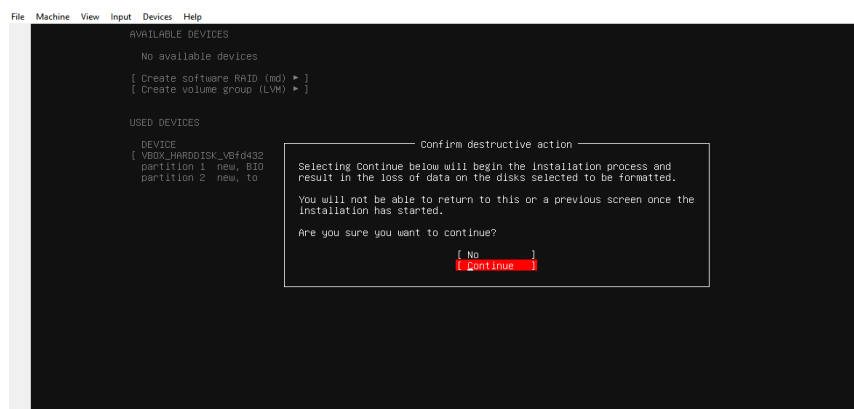
2. Confirm that the partitions belong to the VMware virtual disk.
3. Select Done only after reviewing the complete layout.

Task 12: Confirm the Storage Changes

1. Read the destructive-action warning carefully.
 - Save a screenshot showing the final warning as ubuntu_storage_warning.png.



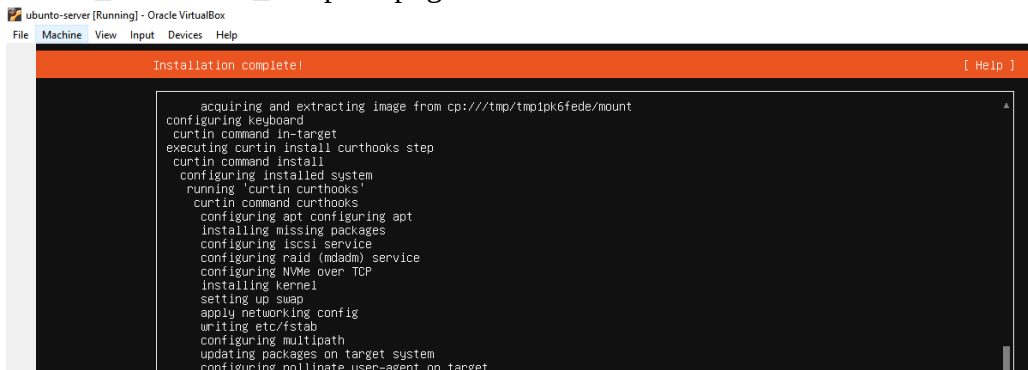
2. Confirm the changes only when you are certain that the VMware virtual disk is selected.
 - Save a screenshot showing the confirmation selection as ubuntu_storage_confirmed.png.



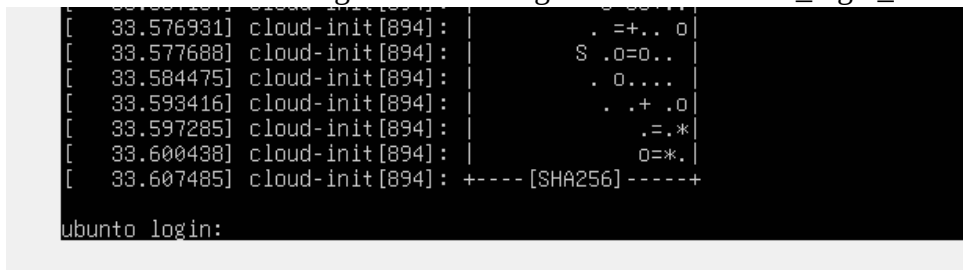
Task 13: Configure the Ubuntu Profile

Complete the Ubuntu profile using the following required values:

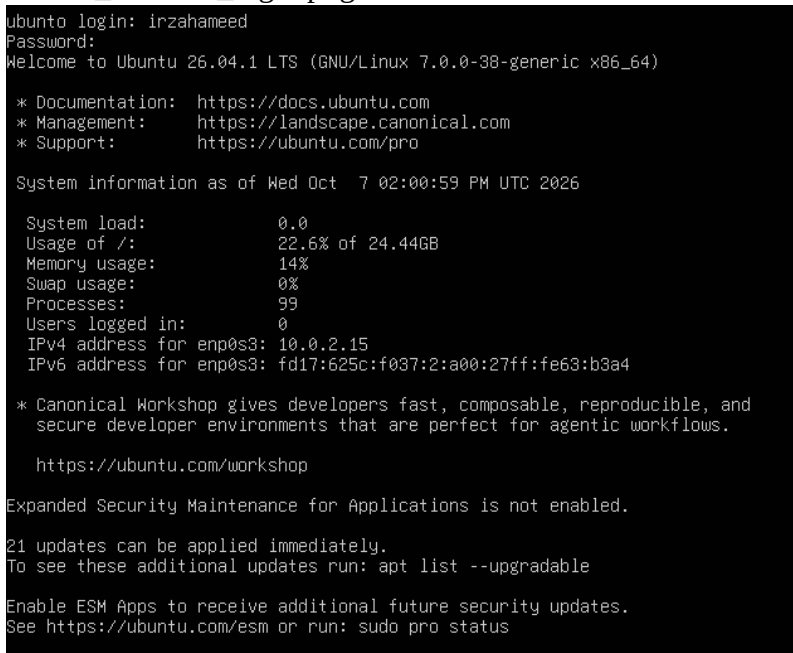
- Save a screenshot of the installation-complete screen as `ubuntu_installation_complete.png`.



2. Disconnect the Ubuntu ISO if VMware asks you to remove the installation media.
 3. Allow the installed Ubuntu Server to start.
- Save a screenshot showing the Ubuntu login screen as `ubuntu_login_screen.png`.



4. Sign in using the Ubuntu username and password configured in Task 13.
- Save a screenshot showing the successful terminal login as `ubuntu_terminal_login.png`.



5. Verify that the terminal prompt follows this format:

`<github-username>@ubuntu:~$`

- Save a clear screenshot showing the complete prompt as `ubuntu_identity_verified.png`.

```

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

irzahameed@ubuntu:~$
irzahameed@ubuntu:~$

```

Task 16: Find the Ubuntu Server IP Address

1. Run the following command inside Ubuntu Server:

ip addr

- Save a screenshot showing the command and its output as `ubuntu_ip_addr_command.png`.

```

irzahameed@ubuntu:~$
irzahameed@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:63:b3:a4 brd ff:ff:ff:ff:ff:ff
    altname enx08002763b3a4
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 83114sec preferred_lft 83114sec
    inet6 fd17:625c:f037:2:a00:27ff:fe63:b3a4/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85951sec preferred_lft 13951sec
    inet6 fe80::a00:27ff:fe63:b3a4/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
irzahameed@ubuntu:~$

```

2. Find the IPv4 address shown after inet for the active network interface.
- Do not use 127.0.0.1; it is the loopback address.
- Save a screenshot clearly showing the correct IPv4 address as `ubuntu_ip_address.png`.

```

    link/ether 08:00:27:63:b3:a4 brd ff:ff:ff:ff:ff:ff
    altname enx08002763b3a4
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 83114sec preferred_lft 83114sec
    inet6 fd17:625c:f037:2:a00:27ff:fe63:b3a4/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85951sec preferred_lft 13951sec
    inet6 fe80::a00:27ff:fe63:b3a4/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
irzahameed@ubuntu:~$

```

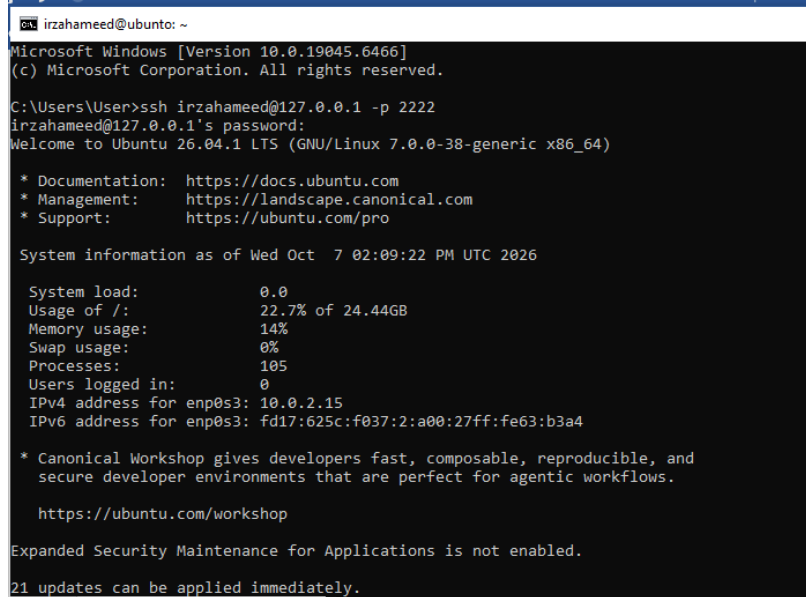
3. Keep your `<github-username>@ubuntu` prompt visible in the screenshot.

Task 17: Connect to Ubuntu Server from Windows

1. Keep the Ubuntu Server virtual machine powered on.
2. Open Command Prompt or PowerShell on Windows.
3. Run the following SSH command, replacing the placeholders with your Ubuntu username and server IP address:

ssh <github-username>@<ubuntu-server-ip>

- Save a screenshot showing the SSH command as windows_ssh_command.png.



```

C:\Users\User>ssh irzahameed@127.0.0.1 -p 2222
irzahameed@127.0.0.1's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Oct  7 02:09:22 PM UTC 2026

System load:          0.0
Usage of /:           22.7% of 24.44GB
Memory usage:         14%
Swap usage:           0%
Processes:            105
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe63:b3a4

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

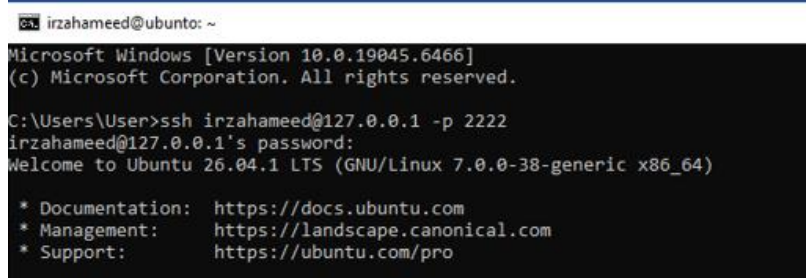
   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.

```

4. Type yes if you are asked to accept the server fingerprint.
- Save a screenshot showing the accepted fingerprint as windows_ssh_fingerprint.png.



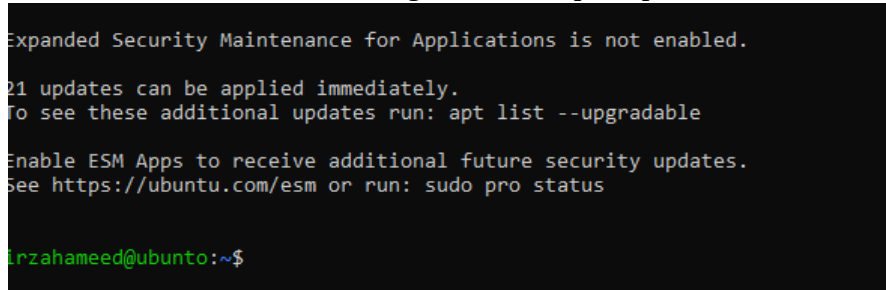
```

C:\Users\User>ssh irzahameed@127.0.0.1 -p 2222
irzahameed@127.0.0.1's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

```

5. Enter your Ubuntu password and complete the remote login.
- The password will not appear while you type; this is normal.
 - Do not expose your password in a screenshot.
 - Save a screenshot showing the successful SSH session as windows_ssh_login.png.
6. Verify that the remote prompt displays <github-username>@ubuntu.
- Save a clear screenshot showing the remote prompt as windows_ssh_identity.png.



```

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

irzahameed@ubuntu:~$

```

If SSH is unavailable, install and start OpenSSH Server inside Ubuntu:

sudo apt update

sudo apt install -y openssh-server

sudo systemctl enable --now ssh